

Different ways to declare variables

1. $x = 2; y = 3$
2. $a = 2$
 $b = 3$
3. $a, b, x, y = 2, 3, 4, 5$

[Here we should give value for all the variable initialized, else it will show error.

For eg:- $a, b, c = 2, 3$ is wrong

$a, b = 2, 3, 4$ is also wrong]

Operator

It is a special symbol used for particular operations between two operands.

For eg:-

$a + b = c$ [Here '+' and '=' are operators. a, b, c are operands.]

Type of operator

1. Arithmetic operator — They are used to perform mathematical operations.
 $+, -, *, /, \%, **, //$

For eg:- $a + b; a - b; a * b; a / b$ etc...

2. Assignment operator — They are used to assign value to variable.
 $=, +=, -=, *=, /=, \%, **, //$

For eg:- $a = 4; b -= 2; c += 3$ etc

3. Relational operator — It is used to compare values and return a Boolean value.

$==, !=, >, <, >=, <=$

For eg:- $n1 > n2$; $n1 \geq n2$; $n1 \neq n2$ etc.

4. logical operators -
and, or, not

A	B	and	or	not-and (NAND)	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

For eg:- $a = \text{True}$
 $\text{not}(a) \Rightarrow \text{output} : \text{True}.$

5. Membership operator - It is used to test whether a value is found within a sequence. It returns a boolean value.

in, not in

For eg:- $\text{qrp} = [1, 2, 3, 4] \Rightarrow \text{output} : \text{False}.$
 9 in qrp

6. Identity operator - It compares the memory address of objects.
is, is not.

7. Bitwise operator - It performs operations on individual bits of integer number.

$\&, |, \sim, \gg, \ll, \wedge$

For eg:- $12 \& 9 \Rightarrow \text{output} : 8$

$12 \rightarrow 1100$

$9 \rightarrow 1001$

AND $1000 \rightarrow 8.$

$\gg \rightarrow$ Shift right operator (Right)

$13 \gg 2 \rightarrow 3$

$\ll \rightarrow$ left side shift operator

$13 \ll 2 \rightarrow 52.$